Updated at 2026.10.27.

Yuntae Ju

Department of Physical Education, Yonsei University
Email: wndbsxo1116@yonsei.ac.kr | Phone: +82-10-8749-8559 |
Website: https://yuntae1116.github.io/

EDUCATION

Yonsei University, Seoul, Korea
M.S. in Exercise Physiology — 2025 - 2027 (expected)
GPA: 4.30/4.30

Yonsei University, Seoul, Korea
B.S. in Physical Education & B.A. in Economics — 2019 - 2023
Graduated in the top 7.7%

MAJOR RESEARCH INTERESTS

Physical Activity Epidemiology · Cardiometabolic Health · Healthy Aging · Wearable & Digital Health · Machine Learning & Bioinformatics

RESEARCH & PROFESSIONAL EXPERIENCE

Researcher, Department of Physical Education, Yonsei University — 2026 - 2027

Bioinformatician, School of Biological Sciences,
Seoul National University — 2026 - Present

Researcher, Frontier Research Institute of Convergence Sports Science,
Yonsei University — 2025 - Present

Active-Duty Officer, Republic of Korea Marine Corps — 2023 - 2025
Honorably discharged as First Lieutenant (Reserves)

Research Assistant, Yonsei University — 2021-2024
Study on the Establishment of an Anti-doping Platform using Sport Big Data Modeling

Reserve Officers' Training Corps (ROTC) Cadet, Republic of Korea — 2021 - 2023

TEACHING EXPERIENCE

Teaching Assistant, Health Education — Spring 2025, Spring 2026

Teaching Assistant, Exercise Physiology — Fall 2025, Fall 2026

PUBLICATIONS

[6] **Ju, Y.**, Lee, J., Kim, D.-H., & Suh, S.-H. (2026). Physical activity phenotypes identified by latent class analysis and their associations with metabolic syndrome among Korean adults. *BMC Public Health*. https://doi.org/10.1186/s12889-026-29957-8

[5] Kim, Y.-H., Lee, J.-H., **Ju, Y.**, & Suh, S.-H. (2026). A review of the effects of aging and exercise on satellite cells and sarcopenia. *Journal of Coaching Development, 28*(2), 116–126. https://doi.org/10.47684/jcd.2026.02.28.2.116

[4] Lee, J.-H., Kim, Y.-H., **Ju, Y.**, & Suh, S.-H. (2026). N-lactoyl-phenylalanine (Lac-Phe): Biosynthetic mechanisms and exercise responses—implications for anti-obesity effects and exercise-based weight management interventions. *Journal of Coaching Development, 28*(2), 87–95. https://doi.org/10.47684/jcd.2026.02.28.2.87

[3] Kim, Y.-H., Lee, J.-H., **Ju, Y.**, Oh, M., & Suh, S.-H. (2025). A review of satellite cell responses to exercise and nutritional interventions. *Journal of Coaching Development, 27*(10), 299–313. https://doi.org/10.47684/jcd.2025.10.27.10.299

[2] Lee, J.-H., Kim, Y.-H., **Ju, Y.**, Oh, M., & Suh, S.-H. (2025). Resistance exercise and protein supplementation strategies to overcome anabolic resistance in aging muscle: A review. *Journal of Coaching Development, 27*(10), 314–328. https://doi.org/10.47684/jcd.2025.10.27.10.314

[1] **Ju, Y.**, Ryoo, H., & Suh, S.-H. (2023). Clustering rookie Korean baseball pitcher and predicting potential by use of machine learning. *Journal of Coaching Development, 25*(2), 190–197. https://doi.org/10.47684/jcd.2023.03.25.2.190

MANUSCRIPTS IN PROGRESS

[2] **Ju, Y.,** & Suh, S.-H. *Artificial Intelligence-Driven Athlete Performance Passport: A New Informatics Framework for Anti-Doping Intelligence.* Drug Testing and Analysis (under review)

[1] **Ju, Y.**, Kim, D.-H., & Suh, S.-H. Artificial Intelligence–Driven Athlete Performance Passport to Enhance Anti-Doping Strategies in Elite Male Weightlifting. *BMC Sports Science, Medicine and Rehabilitation (*submitted)

PATENTS

Method and Apparatus for Determining Doping Using Artificial Intelligence Model. Korean Patent Application No. 10-2026-0208351.

RESEARCH SOFTWARE

MethylDotbogi 2026
Python/R-based bioinformatics tool for automated whole-genome bisulfite sequencing analysis and visualization, including DNA methylation profiling, gene/transposable-element metaplots, and chromosome-wide visualization.

PRESENTATIONS

Ju, Y. & Suh, S-H. (2026). Artificial intelligence-driven athlete performance passport for enhancing anti-doping strategies in elite male weightlifting. 31st Annual Congress of the European College of Sport Science (ECSS), Lausanne, Switzerland.

Xu, Y., **Ju, Y.**, Jung, S., Park, S., Suh, S-H., & Choi, M. H. (2026). Integrating clinical steroid signatures into anti-doping strategies. Cologne Workshop on Doping Analysis, Cologne, Germany.

Ju, Y. & Suh, S-H. (2025). Artificial intelligence-driven athlete performance passport for elite male weightlifting. 67th Sports Medicine Conference, Seoul, Korea.

HONORS & AWARDS

Full Graduate Scholarship (M.S. Program), Yonsei University 2025 - 2027

Jinri Scholarship for Academic Excellence, Yonsei University 2020, 2021, 2022

SKILLS

Programming & Data Analysis: Python, R, STATA, MATLAB, Tableau
Language Skills: Korean (Native), English (Fluent), Japanese (Intermediate)

MATHEMATICS & STATISTICS COURSEWORK

Calculus, Linear Algebra, Mathematical Statistics, Real Analysis, Econometrics, Mathematical Economics, Game Theory, Financial Engineering, Business Analytics, Review Synthesis and Meta-Analysis for Sport Science Research

CERTIFICATION

KBO Official Scorer, Korea Baseball Organization